

	<i>is precise. Will not penalize students if percentage difference is not calculated.</i>	
2(a)(i)	Work is done when a force moves its point of application in the direction of the force.	B1
2(a)(ii)	The mass m is moved without acceleration up a height h, and magnitude of the $F = mg$ Since $E_p = \text{work done by Force } F$ $= \text{force} \times \text{distance moved} \cos \theta$ $= mg \times h \cos 0^\circ$ $= mgh$	M1 M1 M1
2(b)(i)	$\text{Power} = 60 \times 55 \times 9.81 \times 32 / 60$ $= 17300 \text{ W}$	M1 A1
2(b)(ii)1.	$\text{Power} = 1.5 \times 10^4 \times 32 / (60 \sin 45^\circ)$ $= 11300 \text{ W}$	M1 A1
2(b)(ii)2.	$\text{Power input} = (17300 + 11300) \times (100/70)$ $= 40900 \text{ W}$	M1 A1